

## 1.9 Core PyTorch Modules

### 1.9.1 Essential Core Modules

These modules come built-in with PyTorch installation:

```
1 import torch                # Core module - everything you need
2 import torch.autograd       # Automatic differentiation
3 import torch.nn             # Neural network building blocks
4 import torch.optim          # Optimization algorithms
5 import torch.utils.data     # Data loading and processing
6 import torch.jit            # TorchScript for model serialization
7 import torch.distributed    # Distributed training
8 import torch.cuda           # GPU acceleration (NVIDIA)
9 import torch.backends       # Backend selection for training
10 import torch.multiprocessing # Parallel processing
11 import torch.quantization   # Model compression
12 import torch.onnx           # ONNX compatibility
```

Listing 1: Core PyTorch Modules

#### Module Descriptions:

- **torch:** Core module containing all fundamental tensor operations
- **torch.autograd:** Automatic differentiation engine for gradient computation
- **torch.nn:** Neural network components (layers, activation functions, loss functions)
- **torch.optim:** Optimization algorithms (SGD, Adam, RMSprop, etc.)
- **torch.utils.data:** Dataset and DataLoader classes for efficient data handling
- **torch.jit:** TorchScript for model serialization and deployment
- **torch.distributed:** Multi-GPU and multi-machine training
- **torch.cuda:** GPU acceleration utilities
- **torch.backends:** Backend configuration for different hardware
- **torch.multiprocessing:** Parallel processing utilities
- **torch.quantization:** Model compression techniques
- **torch.onnx:** Export models to ONNX format

### 1.9.2 Domain-Specific Libraries

These require separate installation:

```
1 # Computer Vision
2 import torchvision
3 import torchvision.datasets
4 import torchvision.transforms
5 import torchvision.models
6
7 # Natural Language Processing
8 import torchtext
```